

Project Name: Smart Stay

Group: 2

Team Members & Roles:

- **Moaz Gamal Eldin Tawfiq:** Main (Python) , Support (Tableau)
- **Youssef Mohammed Omara:** Main (SQL) , Support (Power BI)
- **Muhammed Gamal Muhammed:** Main (Tableau) , Support (SQL)
- **Nermen Samy Abdelhafez Abozaid:** Main (Excel), Support (Python)
- **Nourhan Saeed Hifny Lashin:** Main (Power BI) , Support (Excel)

Instructor:

Dr. Amal Mahmoud

- **Project Timeline:** 2 Months
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1. Project Planning & Management

- **Project Proposal:** The Smart Stay project is a data analytics initiative focused on the tourism industry. It aims to process large volumes of hotel booking data to examine booking patterns, guest behaviors, and revenue sources. The ultimate goal is to provide crucial intelligence for data-driven decisions and create a scalable framework for a potential national tourism analytics platform in Egypt.
- **Project Plan (2-Month High-Level Timeline):**
 - **Week 1-2: Data Modeling & ETL.** (Youssef, Muhammed)
 - Initial data ingestion, cleaning, and validation in SQL.
 - Design and deployment of the full star schema (10 dimension tables, 1 fact table) .
 - **Week 3-4: Deep Analysis & Feature Engineering.** (Moaz, Nermen)
 - Advanced cleaning, outlier handling (IQR), and feature engineering (e.g., `potential_revenue`, `realized_revenue`) in Python.
 - Exploratory Data Analysis (EDA), correlation analysis, and segmentation.
 - **Week 5-6: BI Dashboard Implementation.** (Muhammed, Nourhan, Moaz, Youssef, Nermen)
 - **Tableau:** Build all worksheets, calculated fields, and assemble the interactive dashboard .
 - **Power BI:** Build the data model, write all DAX measures, and assemble the interactive report .
 - **Excel:** Replicate the workflow using Power Query, Power Pivot, and PivotChart dashboards.
 - **Week 7-8: Finalization & Documentation.** (All)
 - Cross-validation of all KPI metrics between SQL, Python, and BI tools.

- Creation of final documentation (this document) and presentation.
 - **Task Assignment & Roles:** (As listed above).
 - **Risk Assessment & Mitigation Plan:**
 - **Risk:** Duplicated or inconsistent booking records.
 - **Mitigation:** A robust deduplication strategy using `ROW_NUMBER()` was implemented in SQL during the creation of the modeling and guest dimension tables.
 - **Risk:** Data quality issues (e.g., negative ADR, 0 adults with children).
 - **Mitigation:** Strict cleaning rules, imputation (e.g., null children to 0), and filtering were applied in both the SQL and Python phases.
 - **Risk:** Misleading revenue metrics due to 0 values for cancellations.
 - **Mitigation:** Engineered two new features in Python: `potential_revenue` (what would have been earned) and `realized_revenue` (what was earned), providing a clear view of lost revenue.
 - **Risk:** Anomalous data points (e.g., "Non Refund" deposit type having a ~99% cancellation rate).
 - **Mitigation:** The anomaly was identified, documented, and flagged for business-level validation rather than being deleted, preserving data integrity.
 - **Project KPIs (Success Metrics):**
 - **Metric 1:** Deliver three functional, interactive dashboards (Tableau, Power BI, Excel) allowing stakeholders to filter and segment data.
 - **Metric 2:** Achieve 100% data consistency across all platforms, with KPI card values (\$25.24M Revenue, 37.1% Cancel Rate, etc.) matching the master SQL queries.
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2. Literature Review (Problem Context)

This project serves as a key digital innovation initiative for DEPI. The core problem is that raw hotel booking data is voluminous and unstructured, making strategic decision-making difficult. By applying sophisticated analytical methodologies, this project addresses the need for clear insights into booking patterns, revenue optimization, and customer behavior. It establishes a framework for intelligent, data-driven management within Egypt's vital hospitality sector.

3. Requirements Gathering

- **Stakeholder Analysis:**
 - **Primary Stakeholders:** DEPI, Hospitality & Tourism Department decision-makers.
 - **Needs:** Access to strategic insights, identification of market segments, understanding of revenue drivers, and a tool for data-driven policy-making.
- **User Stories & Use Cases:**

- *As a Revenue Manager*, I want to see the **cancellation rate by market segment** and **lead time bin** so I can adjust deposit policies for high-risk bookings.
- *As a Marketing Director*, I want to view **revenue by country** on a map and in a Top 10 list to optimize marketing spend in key markets.
- *As an Operations Manager*, I want to track **ADR by segment** and **revenue by tourism type** to understand the value of different guest types.
- *As an Executive*, I want to see high-level **KPI cards** (Total Revenue, ADR, Cancel Rate) and **yearly performance trends** for a quick overview of business health.
- **Functional Requirements:**
 - The system must load and integrate 11 CSV files (1 fact, 10 dimensions) .
 - The system must provide core calculated metrics: Total Bookings, Total Revenue, Cancellation Rate, Repeat Guest Rate, Avg. ADR, Avg. Lead Time, and Avg. Stay.
 - The dashboards must be filterable by **arrival_date_year**, **hotel**, and **market_segment** .
 - The system must categorize bookings into "Seasons" and "Lead Time Bins" .
- **Non-functional Requirements:**
 - **Accuracy:** All KPI values must be validated against the SQL source of truth.
 - **Usability:** Dashboards must be interactive, with filters and actions enabling cross-filtering and highlighting .
 - **Maintainability:** Code must be well-structured , and the database must use a clean star schema with referential integrity.

4. System Analysis & Design

- **System Architecture:** The project employs a multi-stage analytics architecture:
 - **Data Persistence & Modeling:** A SQL database is used for data cleaning, ETL, and creation of a robust Star Schema .
 - **Deep-Dive Analysis:** Python (with pandas) is used for in-depth statistical analysis, feature engineering, and identifying insights (like lost revenue) not visible in the source data .
 - **Business Intelligence Layer:** Tableau , Power BI , and Excel are used as the user-facing visualization layer, all connecting to the same core data.
- **Database Design (Star Schema):**
 - **Fact Table:** **fact_bookings**. Contains measures (ADR, revenue, lead_time, is_canceled) and foreign keys to all dimensions.
 - **Dimension Tables:**
 - **dim_arrival_date**
 - **dim_booking_type** (Hotel)
 - **dim_customer_type**
 - **dim_deposit_type**
 - **dim_guests_types** (Adults/Children/Babies)
 - **dim_gusts_info** (Guest details, deduplicated by email)

- `dim_marketing` (Segment, Channel, Agent)
- `dim_reservation_status`
- `dim_reservation_details` (Room, Meal)
- `dim_stayed_nights`
- **Data Flow (DFD):**
 - Raw CSV files are loaded into a SQL staging table (`booking_cleaning`).
 - Cleaning, validation, and feature engineering (e.g., `adr_clean`, `full_arrival_date`) are applied .
 - Data is normalized and loaded into the 10 dimension tables .
 - The `booking_modeling` table is joined with all dimensions to populate the final `fact_bookings` table.
 - The cleaned CSVs/Fact table are loaded into Python, Tableau, and Power BI for analysis and visualization.
- **UI/UX Design:** The dashboard layouts are defined by the BI task lists, serving as wireframes.
 - **Structure:** A top-level container for 7 KPI cards. A main body for charts, organized logically (e.g., trends first, then segments, then guest origins) . A dedicated filter pane for global slicers .
- **Technology Stack:**
 - **Database & ETL:** SQL
 - **Analysis & Prototyping:** Python (pandas, numpy, Plotly, JupyterDash)
 - **BI Dashboards:** Tableau , Power BI, and Excel (with Power Query/Pivot).

5. Implementation (Source Code & Execution)

This section summarizes the primary work completed by each team. All code and documentation are intended to be hosted on GitHub.

- **1. SQL (Youssef & Muhammed):**
 - **Phase 1 (Cleaning):** Created a working copy (`booking_cleaning`). Removed duplicates , imputed nulls (e.g., `country` to 'UNK', `children` to 0) , and performed critical ADR validation, flagging and nullifying negative/invalid values .
 - **Phase 2 (Staging):** Created a deduplicated `booking_modeling` table to serve as a stable source for the star schema.
 - **Phase 3 (Dimensions):** Created and populated all 10 dimension tables (e.g., `dim_arrival_date`, `dim_gusts_info`, `dim_marketing`) using surrogate keys . Notably, `dim_gusts_info` was built by partitioning by `email` to create a unique guest profile.
 - **Phase 4 (Fact Table):** Created `fact_bookings` and populated it by joining `booking_modeling` with all dimensions. Applied all foreign key constraints to ensure referential integrity.

- **Phase 5 (Analysis):** Wrote the final insight queries (e.g., Revenue by Season, Cancel % by Lead Time) that serve as the "source of truth" for all dashboards .
- **2. Python (Moaz & Nermen):**
 - **Cleaning & Outliers:** Loaded the dataset and applied advanced outlier handling, notably using the IQR method to identify and cap extreme ADR values .
 - **Feature Engineering:** This was the most critical Python contribution.
 - Created `potential_revenue` (`adr * total_nights`) to represent total booking value.
 - Created `realized_revenue` (0 for canceled bookings) to match the original data's logic.
 - Calculated **"Revenue lost to cancellations": \$16,471,474.30** (a 38.86% loss rate).
 - Created `effective_adr` (`adr * (1 - is_canceled)`) as a revenue yield proxy.
 - **EDA & Analysis:** Performed deep-dive analysis, finding key correlations (e.g., `is_canceled` highest correlation with `lead_time` at +0.29). Analyzed segment performance (e.g., "Online TA" has high ADR but high cancellations). Built a cancellation risk profile.
 - **Dashboarding:** Implemented a full, multi-tab interactive dashboard using **JupyterDash and Plotly** , featuring filters, KPI cards, and all required charts, runnable directly within a notebook .
- **3. Tableau (Muhammed & Moaz):**
 - **Phase 1 (Data Model):** Connected to all CSVs and built the star schema using inner joins in the "Data Source" canvas . Created core calculated fields like `[Cancellation Rate]` , `[Season]` , and `[Lead Time Bins]` . Set geographic roles for `country`.
 - **Phase 2 (Worksheets):** Built all 15+ required worksheets, including 7 KPI text cards , a dual-axis chart for yearly performance , a donut chart for seasonal revenue , bar charts for segments , and a map for revenue by country.
 - **Phase 3 (Dashboard):** Assembled the final dashboard, arranging charts in containers . Added global filters for year, hotel, and segment and applied them to all worksheets. Implemented Dashboard Actions (Filter and Highlight) for full interactivity .
- **4. Power BI (Nourhan & Youssef):**
 - **Phase 1 (Data Model):** Imported all CSVs using "Get Data" . Built the star schema in the "Model" view by creating one-to-many relationships between the dimensions and `fact_bookings`. Set the `country` column's data category to "Country/Region" for mapping.
 - **Phase 2 (DAX):** Wrote all 10+ core DAX measures, including `Total Bookings = COUNT(...)` , `Cancellation Rate = DIVIDE(...)` , and `Avg. ADR = AVERAGE(...)` . Created calculated columns for `Season` and `Lead Time Bins`.

- **Phase 3 (Report):** Built the report page using 7 "Card" visuals for KPIs , a "Line and stacked bar chart" for yearly trends , a "Donut chart" for seasonal revenue , and a "Map" visual for revenue by country.
- **Phase 4 (Interactivity):** Added "Slicer" visuals for year, hotel, and market segment, styling them as dropdowns and buttons . Relied on Power BI's default cross-filtering behavior for interactivity.
- **5. Excel (Nermen & Nourhan):**
 - **Workflow:** As per the project request, the same workflow as Python and SQL was applied.
 - **Data Model:** Data was loaded via Power Query, where cleaning steps (equivalent to SQL Phase 1) were applied. The star schema was built in the Power Pivot Data Model (equivalent to SQL Phase 3-4).
 - **Analysis:** DAX measures (equivalent to Power BI Phase 2) were created in Power Pivot.
 - **Dashboard:** Pivot Tables and Pivot Charts were used to create all required visuals, which were assembled on a single dashboard sheet and controlled by Slicers (equivalent to Power BI Phase 4).

6. Testing & Quality Assurance

- **Test Plan:** The primary testing strategy was cross-platform validation.
 - **Test Case 1 (Critical):** Validate KPI card metrics.
 - **Description:** The values on the Tableau , Power BI, and Excel dashboards must exactly match the output of the master `Smart Stay Insights.sql` queries.
 - **Expected Results:**
 - Total Revenue: \$25.24M
 - Total Bookings: 115,884 (or 115.9K)
 - Cancellation Rate: 37.1%
 - Repeat Guest Rate: 3.2%
 - Avg. ADR: \$101.87
 - **Test Case 2:** Validate dashboard interactivity.
 - **Description:** All slicers (Year, Hotel, Segment) and chart actions (clicking on a bar/pie slice) must filter all other visuals on the page correctly.
- **Bug Reports (Issues Found & Resolved):**
 - **Issue 1:** `revenue` column was misleading as it was 0 for canceled bookings.
 - **Resolution:** Engineered `potential_revenue` in Python to calculate the \$16.47M in lost revenue, providing a critical new insight .
 - **Issue 2:** `Non Refund` deposit type showed a ~99% cancellation rate.
 - **Resolution:** Flagged as a data integrity anomaly for the business to investigate. The data was not altered.
 - **Issue 3:** Guest dimension was fragmented due to multiple bookings from the same email.
 - **Resolution:** Rebuilt `dim_gusts_info` in SQL using `ROW_NUMBER() OVER (PARTITION BY email ...)` to create one unique record per guest email.

- **Issue 4:** Invalid guest records (0 adults, >0 children).
 - **Resolution:** Corrected in the SQL cleaning phase.
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7. Final Presentation & Reports

- **Technical Documentation:** This document serves as the primary technical documentation.
- **User Manual (Dashboard Guide):**
 1. **KPIs:** The cards at the top of the dashboard show the overall health of the business.
 2. **Global Filters:** Use the dropdowns/buttons (Year, Hotel, Market Segment) to filter the entire report.
 3. **Cross-Filtering:** Click on any chart element (e.g., the "Summer" slice of the pie chart or "Portugal" on the map) to filter the entire dashboard by that selection. Click again to clear.
 4. **Tooltips:** Hover over any data point to see more details (e.g., exact revenue, cancel rate).
- **Project Presentation:** The PowerPoint presentation will be the next deliverable, summarizing these findings.